



Suppose that I have an opaque jar with some number of ping pong balls in it. You know that the jar either has 10 ping pong balls in it or 1000 ping pong balls in it. In either case the balls are numbered sequentially. You think that the two possibilities are equally likely.

Suppose that we pick a ball at random. It is #3. Does this favor one hypothesis over the other?

It seems to favor the 10 ping pong ball hypothesis, since that hypothesis makes it more likely that #3 will be picked (rather than some number higher than 10).

Suppose someone offered you 40:1 odds that it is the 10-ball jar — so you win \$40 if it is the 1000 ball jar, and lose \$1 if it is the 10 ball jar.

Should you take the bet?

But how much? How likely should I think that it is that the jar has 10  
ping pong balls in it, vs. 10000?

The answer is not obvious. To answer this we need some rule which will tell us how to calculate probabilities in this kind of case.

What information do we have? We are comparing two different hypotheses: the 10-ball theory and the 1000-ball theory.



Before we drew the ball, we thought that each theory had a 0.5 probability of being correct. This is called the theory's **prior probability**.

We also know something else. We know how likely each theory predicts it to be that we draw the #3 ball.

We know that the 10-ball theory says that there is a 0.1 probability that the #3 ball will be drawn. The 1000-ball theory says that there is 0.001 probability that this ball will be drawn.

This is a claim about what is called **conditional probability**. This is the probability that something will happen if something else is true. What we seem to know here is the probability of the evidence (e) conditional on the hypothesis (h). We write that like this:

*Pr(e|h)*

So we know the prior probability of the hypotheses, and the probability of the evidence given the hypothesis. What we **want** to know is this:

*Pr(h|e)*

This is the probability of the hypothesis given the evidence. It tells us how likely we should take the hypothesis to be, given the evidence we have observed.



How can we do this?

One way to answer these questions employs a widely accepted rule of reasoning called “Bayes’ theorem,” named after Thomas Bayes, an 18th century English mathematician and Presbyterian minister.



To arrive at the theorem, we begin with the following definition of conditional probability:

$$P(a|b) = \frac{P(a \& b)}{P(b)}$$

This says that the probability of  $a$  given  $b$  is the chance that  $a$  and  $b$  are both true divided by the chance that  $b$  is true.

Why might this be true? Intuitively, the idea is this. Take all of the chances that  $b$  could be true, and look at how many of them  $a$  is also true in. Divide the situations in which both  $a$  and  $b$  are true by the total number of situations in which  $b$  is true — and that will tell you how likely it is that  $a$  will be true if  $b$  is. And that just is  $P(a | b)$ .



fine-  
tuning of the  
universe



A solid red circle with a subtle drop shadow, centered on a white background. Inside the circle, the text "Bayes' theorem" is written in a white, sans-serif font, centered horizontally and vertically.

Bayes'  
theorem



the  
argument



objections

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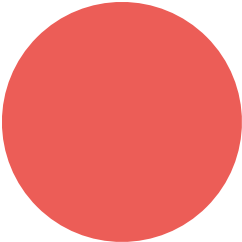








































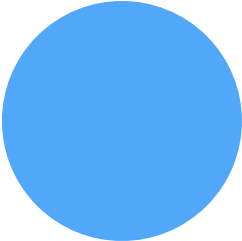




the  
argument



objections













































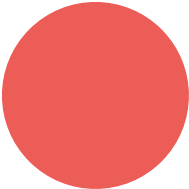












































fine-  
tuning of the  
universe

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Bayes'  
theorem

contemporary physics which will be important to the argument which

Enough about probability (for now). Let's turn now to some results from



follows.

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)



These 'dimensional constants' will be our focus.

energy density of empty space.

One such constant is the cosmological constant, which measures the

The cosmological constant is just a number which (as far as we know)

could have different values consistent with the laws of physics.



One thing that the standard model of physics gives us is a measure of how

(like the cosmological constant) would fall in a certain range.

likely it is, given the laws of the nature, that the fundamental constants

hard to see how life could have evolved. If there were no planets, it is hard

life to have evolved. For example, if there were nothing but hydrogen, it is

We can make certain plausible assumptions about what it would take for

to see how life could have evolved.

constant has a value which would permit the evolution of life.



Given assumptions such as these, we can look at what the standard model

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

This is the claim that the cosmological constant having a life-

supporting value, given the laws of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

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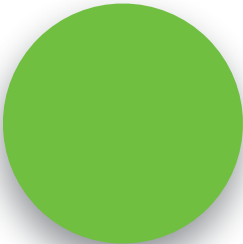
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probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and

in the turn, so we can consider two different hypotheses about the origins

of the universe.

Much as we considered two different hypotheses about the number of balls

hypothesis that it was not.

Let the design hypothesis be the hypothesis that the universe



created by an intelligent designer. Let the non-design hypothesis be the

value in the life-permitting range.

Let  $e$  be the fact that the cosmological constant has a

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e | \text{design}) \equiv 0.5$$



## Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

design hypothesis given evidence is that just plugging in the

With this information in hand, figuring out the probability of the non-

numbers.



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$





$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

It is difficult to think about numbers as large as the denominator of this fraction. But

consider the odds of winning Powerball one trillion times in a row. Call that a “super

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

PowerBar!



that a 'super duper Powerball.'

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Now consider the odds of winning a super Powerball one trillion times in a row.

The odds of this happening are about  $1/10^{44}$  — so much, much higher than the

odd of the universe being life-permitting by chance.

probability of 0.5 to the non-design hypothesis, you should think that the chances of

This means that if you simply take the physics at face value, and begin by assigning a

the non-design hypothesis being true are vastly lower than the chances of winning a



superduperpowerball a trillion times in a row.

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.

minus the very small number we're just discussing:



$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.































2



P

6. *Life exists.*

$$3. P(\text{design}) = P(\text{non-design}) = 0.5.$$

4. Bayes', theorem.

$$\text{C.P.}(\text{God exists}) \approx 1. (7,8)$$



5



P

8. If the universe was created by an

1, 2, 3, 4



independent designer, then God exists.

$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

$$(life/death) \equiv 0.5$$

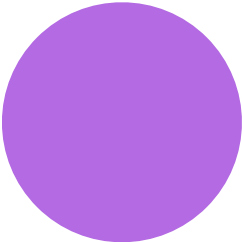
$$(1\text{ife} \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$(\text{design} / \text{life}) \approx 1$$





























Here's an example to illustrate. Suppose that we have

and Ms. C has a 50% chance of winning.



chance of winning, Mr. B has a 30% chance of winning,

three candidates for political office. Mr. A has a 20%

$$P(a|b) = \frac{P(a \& b)}{P(b)}$$